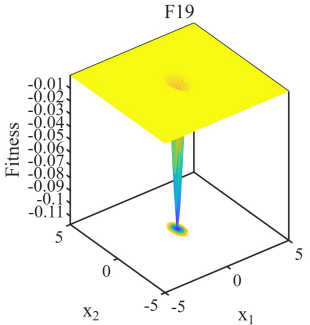
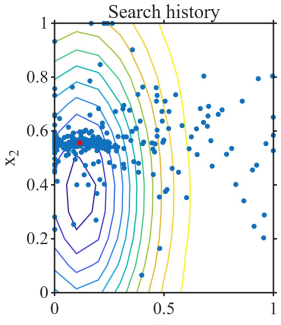
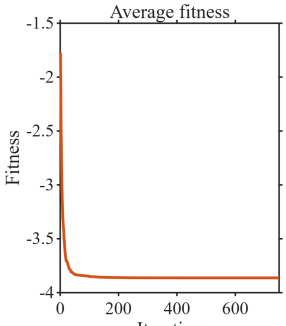
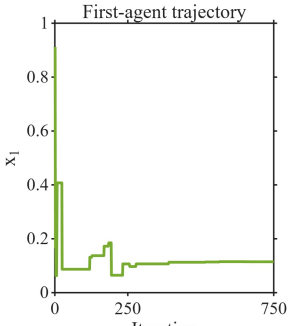
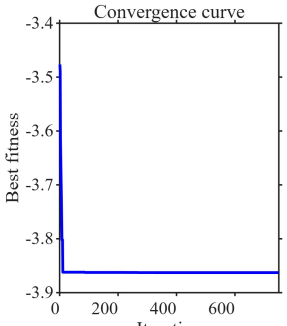

F19
Fitness
-0.01
-0.02
-0.03
-0.04
-0.05
-0.06
-0.07
-0.08
-0.09
-0.1
-0.11
5
0
-5
x2
-5
0
5
x1


Search history
x2
1
0.8
0.6
0.4
0.2
0
0
0.5
1
x1


Average fitness
Fitness
-1.5
-2
-2.5
-3
-3.5
-4
0
200
400
600
Iteration


First-agent trajectory
x1
1
0.8
0.6
0.4
0.2
0
0
250
750
Iteration


Convergence curve
Best fitness
-3.4
-3.5
-3.6
-3.7
-3.8
-3.9
0
200
400
600
Iteration